

```
In [2]: 1 import array
        2 import collections
        3 import copy
        4 import decimal
        5 import math
        6 import pprint
        7 import random
        8 import re
        9 import statistics
       10 import string
       11 import sys
       12 import time
       13 import timeit
       14
       15 import pandas
```

```
In [3]: 1 num1 = 10_00_000
        2 print(num1)
```

```
1000000
```

```
In [4]: 1 print(format(0.1, ".25f"))
```

```
0.10000000000000000055511151
```

```
In [5]: 1 print(format(0.125, ".25f"))
```

```
0.12500000000000000000000000
```

```
In [6]: 1 print(0.1 + 0.1 + 0.1 == 0.3)
```

```
False
```

```
In [7]: 1 print(format(0.3, ".25f"))
```

```
0.29999999999999999888977698
```

```
In [8]: 1 print(format(0.2, ".25f"))
```

```
0.20000000000000000111022302
```

```
In [9]: 1 print(abs((0.1 + 0.1 + 0.1) - 0.3) < 0.001)
```

```
True
```

```
In [10]: 1 print((0.125).as_integer_ratio())
```

```
(1, 8)
```

```
In [11]: 1 print((10).          (25))
```

```
35
```

```
In [12]: 1 a = float(10)
         2 print(a)
```

```
10.0
```

```
In [13]: 1 c = (-4) ** 0.5
         2 print(c)
```

```
(1.2246467991473532e-16+2j)
```

```
In [14]: 1 print(type(c))
```

```
<class 'complex'>
```

```
In [15]: 1 print(c.real)
```

```
1.2246467991473532e-16
```

```
In [16]: 1 print(c.imag)
```

```
2.0
```

```
In [17]: 1 c = 10 + 2j
         2 print(type(c))
```

```
<class 'complex'>
```

```
In [18]: 1 print(c)
```

```
(10+2j)
```

```
In [19]: 1 try:
         2     print("2" < 2)
         3 except TypeError as ex:
         4     print(type(ex).    )
```

```
TypeError
```

```
In [20]: 1 print("2" == 2)
```

```
False
```

```
In [21]: 1 a = 0
         2 b = 1
         3 c = 500
         4 d = 0
         5 e = 1
         6 f = 500
         7 print(id(a), id(b), id(c), id(d), id(e), id(f))
```

```
11370000 11370032 126250629949072 11370000 11370032 126250629947504
```

```
In [22]: 1 print(0.1 * 3 - 0.3 < abs(1e-6))
```

```
True
```

```
In [23]: 1 a = 257
         2 print(a is 257)
```

False

```
<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
/tmp/ipykernel_40114/1958631963.py:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
print(a is 257)
```

```
In [24]: 1 a = 10
         2 b = 0
         3 print(b != 0 and a / b)
```

False

```
In [25]: 1 a = 1e-6
         2 b = 2e-6
         3 tol = 1e-7
         4 print(abs(a - b) <= tol)
```

False

```
In [26]: 1 elapsed = 7835
         2 hours = elapsed // 3600
         3 minutes = (elapsed - (hours * 3600)) // 60
         4 seconds = elapsed - ((hours * 3600) + (minutes * 60))
         5 print(hours, minutes, seconds)
```

2 10 35

```
In [27]: 1 elapsed = 7835
         2 hours = elapsed // (60 * 60)
         3 minutes = (elapsed % (60 * 60)) // 60
         4 seconds = (elapsed % (60 * 60)) % 60
         5 print(hours, minutes, seconds)
```

2 10 35

```
In [28]: 1 a = None
         2 if a == None:
         3     a = "N/A"
         4 print(a)
```

N/A

```
In [29]: 1 a = None
         2 a = "N/A" if a is None else a
         3 print(a)
```

N/A

```
In [30]: 1 score = 700
         2 rating = "Invalid"
         3 if score < 0:
         4     rating = "Negative"
```

```
5 elif score < 580:
6     rating = "Poor"
7 elif score < 670:
8     rating = "Fair"
9 elif score < 740:
10    rating = "Good"
11 elif score < 800:
12    rating = "Very Good"
13 elif score < 850:
14    rating = "Excellent"
15 print(rating)
```

Good

```
In [31]: 1 elapsed = 1000000
2         magnitude = "Invalid"
3         if elapsed < 60:
4             magnitude = "seconds"
5         elif elapsed < (60 * 60):
6             magnitude = "minutes"
7         elif elapsed < (60 * 60 * 24):
8             magnitude = "hours"
9         elif elapsed < (60 * 60 * 24 * 7):
10            magnitude = "days"
11         elif elapsed > (60 * 60 * 24 * 7):
12            magnitude = "weeks"
13         print(magnitude)
```

weeks

```
In [32]: 1 t = [1, 2], 3, 4
2         l = list(t)
3         print(l)
```

[[1, 2], 3, 4]

```
In [33]: 1 l[0][1] = 20
2         print(l)
```

[[1, 20], 3, 4]

```
In [34]: 1 print(t)
```

([1, 20], 3, 4)

```
In [35]: 1 for i in t:
2         print(id(i))
```

126250629147328
11370096
11370128

```
In [36]: 1 for i in l:
2         print(id(i))
```

```
126250629147328
11370096
11370128
```

```
In [37]: 1 for i, j in zip(l, t):
          2     print(i is j)
```

```
True
True
True
```

```
In [38]: 1 s = "ABCDEFGHIIJ"
          2 print(s[3:6:-1])
```

```
In [39]: 1 print(s[3:5 + 1])
```

```
DEF
```

```
In [40]: 1 m1 = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m2 = m1[0:2]
          3 print(m1[0] is m2[1])
```

```
False
```

```
In [41]: 1 m1[0] = "Python"
          2 print(m1)
```

```
['Python', [0, 0, 0], [0, 0, 0]]
```

```
In [42]: 1 print(m2)
```

```
[[0, 0, 0], [0, 0, 0]]
```

```
In [43]: 1 m1[1][0] = 100
```

```
In [44]: 1 print(m1)
```

```
['Python', [100, 0, 0], [0, 0, 0]]
```

```
In [45]: 1 print(m2)
```

```
[[0, 0, 0], [100, 0, 0]]
```

```
In [46]: 1 print("abcdef"[-1:-4])
```

```
In [47]: 1 print("abcdef"[1:4:-1])
```

```
In [48]: 1 print("racecar" == "racecar"[::-1])
```

```
True
```

```
In [49]: 1 list1 = [0, 1, 2, 3, 4]
        2 list1[0:3] = "abc"
        3 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [50]: 1 list1[0:3] = ["a", "b", "c"]
        2 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [51]: 1 list1[0:3] = ["a", "b"]
        2 print(list1)
```

```
['a', 'b', 3, 4]
```

```
In [52]: 1 del list1[2]
        2 print(list1)
```

```
['a', 'b', 4]
```

```
In [53]: 1 del list1[0:2]
        2 print(list1)
```

```
[4]
```

```
In [54]: 1 list1.extend("ABC")
        2 print(list1)
```

```
[4, 'A', 'B', 'C']
```

```
In [55]: 1 list1.insert(0, "Y0!")
        2 print(list1)
```

```
['Y0!', 4, 'A', 'B', 'C']
```

```
In [56]: 1 list1[:-3:-1] = "cb"
        2 print(list1)
```

```
['Y0!', 4, 'A', 'b', 'c']
```

```
In [57]: 1 l = []
        2 print(timeit.timeit("l.append(1)", globals = globals(), number = 100_000))
```

```
0.002109719999907611
```

```
In [58]: 1 print(len(l))
```

```
100000
```

```
In [59]: 1 list2 = []
        2 print(timeit.timeit("list2.insert(0, 1)", globals = globals(), number = 100_000))
```

```
1.7893177260000357
```

```
In [60]: 1 print(len(list2))
```

```
100000
```

```
In [61]: 1 print(1.249788 / 0.0018043)
```

```
692.6719503408524
```

```
In [62]: 1 list2 = []  
2 print(timeit.timeit("list2.insert(len(list2), 1)", globals = globals(), number  
3 = 100_000))
```

```
0.005125395000050048
```

```
In [63]: 1 print(0.0057634 / 0.0018043)
```

```
3.194258161059691
```

```
In [64]: 1 l1 = [[0, 1], 2, "3"]  
2 l2 = l1[:]  
3 for i in l1:  
4     print(id(i))
```

```
126250629156992  
11370064  
11434512
```

```
In [65]: 1 for i in l2:  
2     print(id(i))
```

```
126250629156992  
11370064  
11434512
```

```
In [66]: 1 l2[0][0] = "0"  
2 for i in l1:  
3     print(id(i))
```

```
126250629156992  
11370064  
11434512
```

```
In [67]: 1 for i in l2:  
2     print(id(i))
```

```
126250629156992  
11370064  
11434512
```

```
In [68]: 1 for i in l1:  
2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [69]: 1 for i in l2:
         2     print(i)
```

```
['0', 1]
2
3
```

```
In [70]: 1 l2[1] = "22"
         2 for i in l1:
         3     print(id(i))
```

```
126250629156992
11370064
11434512
```

```
In [71]: 1 for i in l2:
         2     print(id(i))
```

```
126250629156992
126252102803808
11434512
```

```
In [72]: 1 for i in l1:
         2     print(i)
```

```
['0', 1]
2
3
```

```
In [73]: 1 for i in l2:
         2     print(i)
```

```
['0', 1]
22
3
```

```
In [74]: 1 l1 = [[0, 1], 2, "3"]
         2 l2 = l1[:]
         3 l2[0] = ["0", 1]
         4 for i in l1:
         5     print(id(i))
```

```
126250629220160
11370064
11434512
```

```
In [75]: 1 for i in l2:
         2     print(id(i))
```

```
126250629219968
11370064
11434512
```

```
In [76]: 1 for i in l1:
         2     print(i)
```

```
[0, 1]  
2  
3
```

```
In [77]: 1 for i in l2:  
        2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [78]: 1 l1 = [[0, 1], 2, "3"]  
        2 l2 = l1[:]  
        3 print(id(l1), id(l2))
```

```
126250629218368 126250629160320
```

```
In [79]: 1 for i, j in zip(l1, l2):  
        2     print(id(i), id(j))
```

```
126250629218496 126250629218496  
11370064 11370064  
11434512 11434512
```

```
In [80]: 1 for i, j in zip(l1[0], l2[0]):  
        2     print(id(i), id(j))
```

```
11370000 11370000  
11370032 11370032
```

```
In [81]: 1 l1 = [[0, 1], 2, "3"]  
        2 l2 = l1.copy()  
        3 print(id(l1), id(l2))
```

```
126250629221440 126250629222336
```

```
In [82]: 1 for i, j in zip(l1, l2):  
        2     print(id(i), id(j))
```

```
126250629219136 126250629219136  
11370064 11370064  
11434512 11434512
```

```
In [83]: 1 for i, j in zip(l1[0], l2[0]):  
        2     print(id(i), id(j))
```

```
11370000 11370000  
11370032 11370032
```

```
In [84]: 1 l1 = [[0, 1], 2, "3"]  
        2 l2 = copy.deepcopy(l1)  
        3 print(id(l1), id(l2))
```

```
126250629219648 126250629160320
```

```
In [85]: 1 for i, j in zip(l1, l2):  
        2     print(id(i), id(j))
```

```
126250629218496 126250629219968
11370064 11370064
11434512 11434512
```

```
In [86]: 1 for i, j in zip(l1[0], l2[0]):
          2     print(id(i), id(j))
```

```
11370000 11370000
11370032 11370032
11370000
11370032 11370032
```

```
In [87]: 1 t1 = (0, "1", "23")
          2 t2 = t1[:]
          3 print(t1 is t2)
```

```
True
```

```
In [88]: 1 a, b, c, d = "abcd"
          2 print(a, b, c, d)
```

```
a b c d
```

```
In [89]: 1 s = "FfEeDdCcBbAa"
          2 lowerLetters = s[::-2]
          3 upperLetters = s[-2::-2]
          4 print(lowerLetters)
```

```
abcdef
```

```
In [90]: 1 print(upperLetters)
```

```
ABCDEF
```

```
In [91]: 1 t1 = 1, 2, 3, 4, 5, 6
          2 t2 = 7, 8, 9, 10
          3 t3 = 11, 12, 13, 14, 15, 16, 17
          4 t = t1 + t2 + t3
          5 l = list(t)
          6 l[::2] = [0] * len(l[::2])
          7 print(tuple(l))
```

```
(0, 2, 0, 4, 0, 6, 0, 8, 0, 10, 0, 12, 0, 14, 0, 16, 0)
```

```
In [92]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m[0][0] = 1
          3 m[1][1] = 1
          4 m[2][2] = 1
          5 print(m)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

```
In [93]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m1 = copy.deepcopy(m)
          3 m1[0][0] = 1
```

```
4 m1[1][1] = 1
5 m1[2][2] = 1
6 print(m1)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

In [94]:

```
print(m)
1
2 data = [(100, "USD", "EUR", 0.83), (100, "USD", "CAD", 1.27), (100, "CAD",
3 "EUR", 0.65)]
4 USDAmountInEUR = data[0][0] * data[0][3]
5 print(str(data[0][0]) + " " + data[0][1] + " = " + str(USDAmountInEUR) + " " +
str(data[0][2]))
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
100 USD = 83.0 EUR
```

In [95]:

```
1 USDAmountInCAD = data[1][0] * data[1][3]
2 print(str(data[1][0]) + " " + data[1][1] + " = " + str(USDAmountInCAD) + " " +
str(data[1][2]))
```

```
100 USD = 127.0 CAD
```

In [96]:

```
1 CADAmountInEUR = data[2][0] * data[2][3]
2 print(str(data[2][0]) + " " + data[2][1] + " = " + str(CADAmountInEUR) + " " +
str(data[2][2]))
```

```
100 CAD = 65.0 EUR
```

In [97]:

```
1 print(ord("A"))
```

```
65
```

In [98]:

```
1 print("\N{SNAKE}")
```



In [99]:

```
1 print(ord("\N{SNAKE}"))
```

```
128013
```

In [100]:

```
1 print(hex(ord("\N{SNAKE}")))
```

```
0x1f40d
```

In [101]:

```
1 print("\u03b1")
```

```
α
```

In [102]:

```
1 print("\U0001f40d")
```



In [103]:

```
1 print(int("1f40d", 16))
```

```
128013
```

```
In [104]: 1 print("\N[SNAKE] is a snake")
```

```
🐍 is a snake
```

```
In [105]: 1 print(int("7ff", 16))
```

```
2047
```

```
In [106]: 1 print(int("ffff", 16))
```

```
65535
```

```
In [107]: 1 print(int("10ffff", 16))
```

```
1114111
```

```
In [108]: 1 print(bin(2047))
```

```
0b1111111111
```

```
In [109]: 1 print(len(bin(2047)))
```

```
13
```

```
In [110]: 1 print(bin(65535))
```

```
0b1111111111111111
```

```
In [111]: 1 print(len(bin(65535)))
```

```
18
```

```
In [112]: 1 print(bin(1114111))
```

```
0b10000111111111111111
```

```
In [113]: 1 print(len(bin(1114111)))
```

```
23
```

```
In [114]: 1 print(bin(128013))
```

```
0b11111010000001101
```

```
In [115]: 1 print("\n{snake}")
```

```
{snake}
```

```
In [116]: 1 a = True  
2 b = True  
3 print(a is b)
```

```
True
```

```
In [117]: 1 a = False  
2 b = False
```

```
3 print(a is b)
```

```
True
```

```
In [118]: 1 a = True
          2 print(id(a))
```

```
11120192
```

```
In [119]: 1 a = False
          2 print(id(a))
```

```
11045056
```

```
In [120]: 1 b = False
          2 print(id(b))
```

```
11045056
```

```
In [121]: 1 ?
```

Docstring:

Return the lowest index in S where substring sub is found, such that sub is contained within S[start:end].

Optional arguments start and end are interpreted as in slice notation.

Return -1 on failure.

Type: method_descriptor

```
In [122]: 1 print(f"1 + 1 = {1 + 1}")
```

```
1 + 1 = 2
```

```
In [123]: 1 bid = 1.5760
          2 ask = 1.5763
          3 print("bid: {}, ask: {}, spread: {}".format(bid, ask, ask - bid))
```

```
bid: 1.576, ask: 1.5763, spread: 0.000299999999999996696
```

```
In [124]: 1 print("bid: {:.4f}, ask: {:.4f}, spread: {:.4f}".format(bid, ask, ask - bid))
```

```
bid: 1.5760, ask: 1.5763, spread: 0.0003
```

```
In [125]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {ask - bid:.4f}")
```

```
bid: 1.5760, ask: 1.5763, spread: 0.0003
```

```
In [126]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {(ask - bid):.4f}")
```

```
bid: 1.5760, ask: 1.5763, spread: 0.0003
```

```
In [127]: 1 string1 = "β, β, #"
          2 string1 = "Π, υ, θ, ω, ν"
          3 chars = string1.split(", ")
          4 print(chars)
```

```
['Π', 'υ', 'θ', 'ω', 'ν']
```

```
In [128]: 1 list1 = []
          2 for c in chars:
          3     list1.append(hex(ord(c)))
          4 print(list1)
```

```
['0x3a0', '0x3cd', '0x3b8', '0x3c9', '0x3bd']
```

```
In [129]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.upper())
          4 print(list1)
```

```
['Π', 'Υ', 'Θ', 'Ω', 'Ν']
```

```
In [130]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.lower())
          4 print(list1)
```

```
['π', 'υ', 'θ', 'ω', 'ν']
```

```
In [131]: 1 a = 42
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 42 is even
```

```
In [132]: 1 a = 41
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 41 is not even
```

```
In [133]: 1 a = 42
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 42 is even
```

```
In [134]: 1 a = 41
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 41 is not even
```

```
In [135]: 1 a = 0.3
          2 b = 0.1
          3 print(f"{a} / {b} = {a / b:.4f}")
```

```
0.3 / 0.1 = 3.0000
```

```
In [136]: 1 a = 42
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

The number 42 is even

```
In [137]: 1 a = 421
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

The number 421 is not even

```
In [138]: 1 print(range(10))
```

range(0, 10)

```
In [139]: 1 print(type(range(10)))
```

<class 'range'>

```
In [140]: 1 for i in range(3):
          2     print(i)
          3     print(i)
```

0
1
2
2

```
In [141]: 1 for i in "abc":
          2     pass
          3     print(i)
```

c

```
In [142]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
          2 for r in range(len(raggedArray)):
          3     for c in range(len(raggedArray[r])):
          4         print(f"[{r}, {c}] = {raggedArray[r][c]}")
          5     print("-" * 5)
```

[0, 0] = 1
[0, 1] = 2

[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6

[2, 0] = 7
[2, 1] = 8
[2, 2] = 9

[3, 0] = 10

```
In [143]: 1 identityMatrix = []
          2 size = 10
          3 for r in range(size):
          4     row = []
          5     for c in range(size):
```

```

6         if r == c:
7             row.append(1)
8         else:
9             row.append(0)
10        identityMatrix.append(row)
11    identityMatrix

```

```

Out[143]: [[1, 0, 0, 0, 0, 0, 0, 0, 0, 0],
           [0, 1, 0, 0, 0, 0, 0, 0, 0, 0],
           [0, 0, 1, 0, 0, 0, 0, 0, 0, 0],
           [0, 0, 0, 1, 0, 0, 0, 0, 0, 0],
           [0, 0, 0, 0, 1, 0, 0, 0, 0, 0],
           [0, 0, 0, 0, 0, 1, 0, 0, 0, 0],
           [0, 0, 0, 0, 0, 0, 1, 0, 0, 0],
           [0, 0, 0, 0, 0, 0, 0, 1, 0, 0],
           [0, 0, 0, 0, 0, 0, 0, 0, 1, 0],
           [0, 0, 0, 0, 0, 0, 0, 0, 0, 1]]

```

```

In [144]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
          2 for i, r in enumerate(raggedArray):
          3     for j, e in enumerate(raggedArray[i]): # e for element, cuz it's not a
          4         column technically
          5             print(f"[{i}, {j}] = {raggedArray[i][j]}")
              print("-" * 8)

```

```

[0, 0] = 1
[0, 1] = 2
-----
[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6
-----
[2, 0] = 7
[2, 1] = 8
[2, 2] = 9
-----
[3, 0] = 10
-----

```

```

In [145]: 1 print(f"{0.1:.4f}")

```

```
0.1000
```

```

In [146]: 1 print("{:.4f}".format(0.1))

```

```
0.1000
```

```

In [147]: 1 total, count = (0,) * 2
          2 data = [10.5, 11.2, 9.8, None, 11.5, None]
          3 for e in data:
          4     if e:
          5         total += e
          6         count += 1
          7 average = total / count
          8 print(average)

```

10.75

```
In [148]: 1 for i, e in enumerate(data):
          2     if not e:
          3         data[i] = average
          4 print(data)
```

[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]

```
In [149]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
          2 count = sum(1 for e in data if e)
          3 total = sum(e for e in data if e)
          4 average = total / count
          5 print(sum, average)
```

<built-in function sum> 10.75

```
In [150]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
          2 average = statistics.fmean(e for e in data if e)
          3 data = [average if e is None else e for e in data]
          4 print(data)
```

[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]

```
In [151]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
          2 df = pandas.DataFrame(data)
          3 df.fillna(df.mean(), inplace = True)
          4 print(df[0].tolist())
```

[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]

```
In [152]: 1 print(len(str((1, 2))))
```

6

```
In [153]: 1 print(str((1, 2)))
```

(1, 2)

```
In [154]: 1 m, n = 4, 5
          2 for i in range(1, m):
          3     for j in range(1, n):
          4         print(f"{i} x {j} = {i * j}")
          5     print("-" * 8)
```

```
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
-----
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
-----
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
-----
```

In [155]:

```
1
2
3 data = ("2021-01-01", 10, 20], ["2021-01-02", 20, 18], ["2021-01-03", -10,
4 10], ["2021-01-04", 100, 102], ["2021-01-05", 20, 45])
5 largest = 0
6 for e in data:
7     distance = abs(e[2] - e[1])
8     e.append(distance)
9     if distance > largest:
10         dateOfLargest = e[0]
11         largest = distance
12 print(data)
13 print(dateOfLargest)
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```
In [157]: 1 tuple1 = ([0],)
          2 print(type(tuple1))
          3 print(tuple1)
          4 print(id(tuple1))
          5 print(id(tuple1[0]))
          6 tuple1[0][0] += 1
          7 print(tuple1)
          8 print(id(tuple1))
          9 print(id(tuple1[0]))
```

```
<class 'tuple'>
([0],)
126250629919808
126250629056256
([1],)
126250629919808
126250629056256
```

```
In [158]: 1 print(hash(True))
          2 a = True
          3 print(hash(a))
          4 b = True
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
1
1
1
1
```

```
In [159]: 1 print(hash(257))
          2 a = 257
          3 print(hash(a))
          4 b = 257
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
257
257
257
257
```

```
In [160]: 1 print(hash(2570))
          2 a = 2570
          3 print(hash(a))
          4 b = 2570
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
2570
2570
2570
2570
```

```
In [161]: 1 print(hash(25))
          2 a = 25
          3 print(hash(a))
          4 b = 25
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
25
25
25
25
```

```
In [162]: 1 print(hash(2.5))
          2 a = 2.5
          3 print(hash(a))
          4 b = 2.5
          5 print(hash(b))
          6 print(hex(hash(b)))
          7 a = b
          8 print(len(hex(hash(a))))
          9 print(hash(b))
```

```
1152921504606846978
1152921504606846978
1152921504606846978
0x10000000000000002
18
1152921504606846978
```

```
In [163]: 1 for i in range(5):
          2     for j in range(4):
          3         pass
          4 print(i, j)
```

```
4 3
3
```

```
In [164]: 1 s1 = """
          2 "'And' and 'or' are the basic operations of logic. Together with 'no' (the
          3 logical operation
          4 of negation) they are a complete set of basic logical operations – all other
          5 logical
          6 operations, no matter how complex, can be obtained by suitable combinations of
          7 these."
          8 — John von Neumann, The Computer and the Brain
          9 """
          10
          11 s2 = """
          12 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
```

```

7   incididunt ut
8   labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud
9   exercitation ullamco laboris
10  nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in
11  reprehenderit in voluptate velit
12  esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat
13  non proident, sunt
14  in culpa qui officia deserunt mollit anim id est laborum.
15  """
16
17  charCounts = {}
18  for c in s1:
19      if (ord(c) >= ord("A") and ord(c) <= ord("Z")) or (ord(c) >= ord("a")
20      and ord(c) <= ord("z")):
21          charCounts[c] = charCounts.get(c, 0) + 1
22  print(charCounts)

```

```

{'A': 1, 'n': 20, 'd': 4, 'a': 23, 'o': 31, 'r': 12, 'e': 27, 't': 21, 'h': 11, 'b': 7, 's': 9, 'i': 17, 'c': 10, 'p': 7, 'f': 4, 'l': 12, 'g': 6, 'T': 2, 'w': 2, 'y': 2, 'm': 6, 'x': 1, 'u': 3, 'J': 1, 'v': 1, 'N': 1, 'C': 1, 'B': 1}

```

```

In [165]: 1 d1 = {"a": 10, "b": 20, "c": 30}
2 d2 = {"d": 100, "e": 200, "f": 300}
3 d3 = {"f": 30, "g": 40}
4 keysList, valuesList = [], []
5 for d in d1, d2, d3:
6     for k, v in d.items():
7         keysList.append(k)
8         valuesList.append(v)
9 print(keysList)
10 print(valuesList)

```

```

['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
[10, 20, 30, 100, 200, 300, 30, 40]

```

```

In [166]: 1 keysList, valuesList = [], []
2 keysList = list(d1.keys())
3 keysList.extend(list(d2.keys()))
4 keysList.extend(list(d3.keys()))
5 print(keysList)

```

```
['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
```

```
In [167]: 1 valuesList = list(d1.values())
          2 valuesList.extend(list(d2.values()))
          3 valuesList.extend(list(d3.values()))
          4 print(valuesList)
```

```
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [168]: 1 print(list(d1.keys()) + list(d2.keys()) + list(d3.keys()))
          2 print(list(d1.values()) + list(d2.values()) + list(d3.values()))
```

```
['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
```

```
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [169]: 1 grades = {"John": [90, 95, 98], "Eric": [86, 84, 92], "Michael": [90, 89, 85]}
          2 exam = {"Eric": 99, "John": 100}
          3 for k, v in grades.items():
          4     if k in exam:
          5         v.insert(0, exam[k])
          6     else:
          7         v.insert(0, None)
          8 print(grades)
```

```
{'John': [100, 90, 95, 98], 'Eric': [99, 86, 84, 92], 'Michael': [None, 90, 89, 85]}
```

```
In [170]: 1 for k, v in grades.items():
          2     score = exam.get(k)
          3     grades[k] = [score] + grades[k]
          4 print(grades)
```

```
{'John': [100, 100, 90, 95, 98], 'Eric': [99, 99, 86, 84, 92], 'Michael': [None, None, 90, 89, 85]}
```

```
In [171]: 1 for k, v in grades.items():
          2     v.insert(0, exam.get(k))
          3 print(grades)
```

```
{'John': [100, 100, 100, 90, 95, 98], 'Eric': [99, 99, 99, 86, 84, 92], 'Michael': [None, None, None, 90, 89, 85]}
```

```
In [172]: 1 for k in grades:
          2     print(k)
```

```
John
Eric
Michael
```

```
Eric
Michael
```

```
In [173]: 1 l = ["AAPL", "AAPL", "Aapl", "aapl", "MSFT"]
          2 print(list(set(l)))
```

```
['MSFT', 'aapl', 'Aapl', 'AAPL']
```

```
In [174]: 1 l = [e.casefold() for e in l]
          2 print(list(set(l)))
```

```
['aapl', 'msft']
```

```
In [175]: 1 se = {e.casefold() for e in l}
          2 print(se)
```

```
{'aapl', 'msft'}
```

```
In [176]: 1 data = {"d1": {"a": 1, "b": 2, "c": 3}, "d2": {"b": 20, "c": 30, "d": 40},
          2 "d3": {"d": 100, "x": 200}}
          3 uniqueKeys = {k for v in (data["d1"], data["d2"], data["d3"]) for k in v}
          print(uniqueKeys)
```

```
{'b', 'c', 'a', 'd', 'x'}
```

```
In [177]: 1 try:
          2     s1 = {[0, 1]}
          3     print(s1)
          4 except TypeError as ex:
          5     print(type(ex).name, ":", " ", *ex.args, ".", sep = " ")
```

```
TypeError: cannot use 'list' as a set element (unhashable type: 'list').
```

```
In [178]: 1 m = [0] * 3
          2 print(m)
          3 print(m[0] is m[1])
```

```
[0, 0, 0]
True
```

```
In [179]: 1 m = [[0] * 3 for _ in range(3)]
          2 print(m)
          3 print(m[0] is m[1])
          4 print(m[0][0] is m[0][1])
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
False
True
```

```
In [180]: 1 m = [[10000 for i in range(3)] for _ in range(4)]
          2 print(m)
```

```
[[10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000]]
```

```
In [181]: 1 pprint.pprint(m, width = 1)
```

```
[[10000,
 10000,
 10000],
 [10000,
 10000,
 10000],
 [10000,
 10000,
 10000],
 [10000,
 10000,
 10000]]
```

```
In [182]: 1 print(m[0] is m[1])
          2 print(m[0][0] is m[0][1])
          3 print(hash(m[0][0]), hash(m[0][1]))
```

```
False
True
10000 10000
```

```
In [183]: 1 print(id(m[0][0]), id(m[0][1]))
```

```
126250629275408 126250629275408
```

```
In [184]: 1 m[0][0] = 1001
          2 pprint.pprint(m)
```

```
[[1001, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000]]
```

```
In [185]: 1 print(id(m[0][0]), id(m[0][1]))
          2 print(m[0][0] is m[0][1])
```

```
126250629271536 126250629275408
False
```

```
In [186]: data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
1 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
2 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
3 ranges = []
4 for d in data:
5     ranges.append(d["high"] - d["low"])
print(ranges)
```

```
[30, 50, 70, 90]
```

```
In [187]: data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
1 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
2 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
3 ranges = [d["high"] - d["low"] for d in data]
print(ranges)
```

```
[30, 50, 70, 90]
```

In [188]:

```
1 bogusComposites = {i for i in range(1, 101) for j in range(2, 10) if i % j == 0}
2 allNumbers = {i for i in range(11, 101)} | {1}
3 print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

In [189]:

```
1 allNumbers = set(range(1, 101))
2 print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

In [190]:

```
1 data = [{"symbol": "ABCD", "name": "ABCD Company", "ranking": 2, "risk": 0.2},
2         {"symbol": "BCDE", "name": "BCDE Company", "ranking": 5, "risk": 0.2},
3         {"symbol": "CDEF", "name": "CDEF Company", "ranking": 8, "risk": 0.5},
         {"symbol": "DEFG", "name": "DEFG Company", "ranking": 7, "risk": 0.8},
         {"symbol": "EFGH", "name": "EFGH Company", "ranking": 9, "risk": 0.6},
         {"symbol": "FGHI", "name": "FGHI Company", "ranking": 10, "risk": 0.1},
         {"symbol": "GHIJ", "name": "GHIJ Company", "ranking": 3, "risk": 0.6},
         {"symbol": "HIJK", "name": "HIJK Company", "ranking": 5, "risk": 0.5},
         {"symbol": "IJKL", "name": "IJKL Company", "ranking": 5, "risk": 0.7},
         {"symbol": "JKLM", "name": "JKLM Company", "ranking": 6, "risk": 0.9},
         {"symbol": "KLMN", "name": "KLMN Company", "ranking": 6, "risk": 0.4},
         {"symbol": "LMNO", "name": "LMNO Company", "ranking": 8, "risk": 0.4},
         {"symbol": "MNOP", "name": "MNOP Company", "ranking": 8, "risk": 0.2},
         {"symbol": "NOPQ", "name": "NOPQ Company", "ranking": 1, "risk": 0.5},
         {"symbol": "OPQR", "name": "OPQR Company", "ranking": 9, "risk": 0.2},
         {"symbol": "PQRS", "name": "PQRS Company", "ranking": 10, "risk": 0.9},
         {"symbol": "QRST", "name": "QRST Company", "ranking": 3, "risk": 0.4},
         {"symbol": "RSTU", "name": "RSTU Company", "ranking": 7, "risk": 0.3},
         {"symbol": "STUV", "name": "STUV Company", "ranking": 8, "risk": 0.1},
         {"symbol": "TUVW", "name": "TUVW Company", "ranking": 7, "risk": 0.9}]
filtered = [d for d in data if d["ranking"] > 5 and d["risk"] < 0.6]
print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5}, {'symbol': 'FGHI', 'name': 'FGHI Company', 'ranking': 10, 'risk': 0.1}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking': 6, 'risk': 0.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4}, {'symbol': 'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2}, {'symbol': 'OPQR', 'name': 'OPQR Company', 'ranking': 9, 'risk': 0.2}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7, 'risk': 0.3}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1}]
```

In [191]:

```
1 for d in filtered:
2     d["weight"] = float(f"{d["ranking"] * d["risk"]:.2f}")
3 print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5, 'weight': 4.0}, {'symbol': 'FGHI', 'name': 'FGHI Company', 'ranking': 10, 'risk': 0.1, 'weight': 1.0}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking': 6, 'risk': 0.4, 'weight': 2.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4, 'weight': 3.2}, {'symbol': 'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2, 'weight': 1.6}, {'symbol': 'OPQR', 'name': 'OPQR Company', 'ranking': 9, 'risk': 0.2, 'weight': 1.8}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7, 'risk': 0.3, 'weight': 2.1}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1, 'weight': 0.8}]
```

```
In [192]: 1 weightedStocks = [dict(symbol = d["symbol"], weight = d["weight"]) for d in
          2 filtered]
          print(weightedStocks)
```

```
[{'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [193]: 1 weightedStocks = [dict(symbol = d["symbol"], weight = round(d["ranking"] *
          2 d["risk"], 2)) for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]
          print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [194]: 1 weightedStocks = [{"symbol": d["symbol"], "weight": round(d["ranking"] *
          2 d["risk"], 2)} for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]
          print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [195]: 1 m = [[0] * 2] * 3
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
```

```
[[0, 0], [0, 0], [0, 0]]
11370000
11370000
11370000
11370000
11370000
11370000
```

```
In [196]: 1 m = [{0}] * 2] * 3
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
126250629090432
126250629090432
126250629090432
126250629090432
126250629090432
126250629090432
```

```
In [197]: 1 m[0][0] = 1
          2 for r in range(len(m)):
```

```
3         for c in range(len(m[r])):
4             print(id(m[r][c]))
5     print(m)
```

```
11370032
126250629090432
11370032
126250629090432
11370032
126250629090432
[[1, {0}], [1, {0}], [1, {0}]]
```

```
In [198]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
2         print(m)
3         for r in range(len(m)):
4             for c in range(len(m[r])):
5                 print(id(m[r][c]))
6         m[0][0] = 1
7         print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
126250629091552
126250629092224
126250629088192
126250629091328
126250629089536
126250629089312
[[1, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [199]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
2         print(m)
3         for r in range(len(m)):
4             for c in range(len(m[r])):
5                 print(id(m[r][c]))
6         m[0][0] = {1}
7         print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
126250629091552
126250629090432
126250629092672
126250629092000
126250629091776
126250629093344
[[{1}, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [200]: 1 m = [[0 for _ in range(2)] for _ in range(3)]
2         for r in range(len(m)):
3             for c in range(len(m[r])):
4                 print(id(m[r][c]))
```

```
11370000
11370000
11370000
11370000
11370000
11370000
```

```
In [201]: 1 m = [[10000 for _ in range(2)] for _ in range(3)]
2         for r in range(len(m)):
3             for c in range(len(m[r])):
4                 print(id(m[r][c]))
5         m[0][0] = 1
6         print(m)
```

```
126250629276976
126250629276976
126250629276976
126250629276976
126250629276976
126250629276976
[[1, 10000], [10000, 10000], [10000, 10000]]
```

```
In [202]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
2         for r in range(len(m)):
3             for c in range(len(m[r])):
4                 print(id(m[r][c]))
```

```
126250629090432
126250629090880
126250629088192
126250629091328
126250629093344
126250629087968
```

```
In [203]: 1 # print(input("Text box displayed ?"))
```

```
In [204]: 1 a = 0
2         b = complex(real = 3.0, imag = 4)
3         print(repr(a))
4         print(repr(b))
5         print(b)
```

```
0
(3+4j)
(3+4j)
```

```
In [205]: 1 values = [4, 3, 1, 2]
2         try:
3             minimum = abs(values[0])
4             for value in values[1:]:
5                 if abs(value) < minimum:
6                     minimum = abs(value)
7         except IndexError:
8             minimum = 0
9         pass
10        finally:
11            print(f"Minimum is: {minimum}")
```

```
Minimum is: 1
```

```
In [206]: 1 values = []
2
3 try:
4     minimum = abs(values[0])
5     for value in values[1:]:
6         if abs(value) < minimum:
7             minimum = abs(value)
8 except IndexError:
9     minimum = 0
10 pass
11 finally:
12     print(f"Minimum is: {minimum}")
```

Minimum is: 0

```
In [58]: 1 try:
2         raise ValueError("Custom message")
3 except ValueError as ex:
4     print(type(ex).      + ": " + ex.args[0] + ".")
```

ValueError: Custom message.

```
In [208]: 1 l = [4, 3, 2, 1, 0]
2 iterator = iter(l)
3 try:
4     while True:
5         print(next(iterator))
6 except StopIteration as ex:
7     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
4
3
2
1
0
StopIteration: .
```

```
In [209]: 1 try:
2         raise IndexError
3 except IndexError as ex:
4     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

IndexError: .

```
In [210]: 1 rangeObject = range(10)
2 rangeIterator = iter(rangeObject)
3 print(next(rangeIterator))
4 print(next(rangeIterator))
5 print(next(rangeIterator))
6 print(list(rangeIterator))
```

```
0
1
2
[3, 4, 5, 6, 7, 8, 9]
```

```
In [211]: 1 rangeObj = range(5)
          2 print(list(rangeObj))
          3 print(list(rangeObj))
```

```
[0, 1, 2, 3, 4]
[0, 1, 2, 3, 4]
```

```
In [212]: 1 enumIterator = enumerate(string.ascii_lowercase)
          2 print(enumIterator)
          3 l = list(enumIterator)
          4 print(l)
          5 l = list(enumIterator)
          6 print(dir(type(iter([0]))))
          7 print(l)
          8 try:
          9     print(next(enumIterator))
         10 except StopIteration as ex:
         11     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
<enumerate object at 0x72d303e88fe0>
[(0, 'a'), (1, 'b'), (2, 'c'), (3, 'd'), (4, 'e'), (5, 'f'), (6, 'g'), (7, 'h'), (8, 'i'), (9, 'j'),
(10, 'k'), (11, 'l'), (12, 'm'), (13, 'n'), (14, 'o'), (15, 'p'), (16, 'q'), (17, 'r'), (18, 's'),
(19, 't'), (20, 'u'), (21, 'v'), (22, 'w'), (23, 'x'), (24, 'y'), (25, 'z')]
['_class_', '__delattr__', '__dir__', '__doc__', '__eq__', '__format__', '__ge__',
'__getattr__', '__getstate__', '__gt__', '__hash__', '__init__', '__init_subclass__',
'__iter__', '__le__', '__length_hint__', '__lt__', '__ne__', '__new__', '__next__', '__reduce__',
'__reduce_ex__', '__repr__', '__setattr__', '__setstate__', '__sizeof__', '__str__',
'__subclasshook__']
[]
StopIteration: .
```

```
In [213]: 1 print(type(enumIterator))
          2 print(type(iter(enumIterator)))
          3 print(issubclass(type(iter([0])), type(enumIterator)))
```

```
<class 'enumerate'>
<class 'enumerate'>
False
```

```
In [214]: 1 print(iter(enumIterator) is enumIterator)
```

```
True
```

```
In [215]: 1 print(type(iter({0: 1, 2})))
```

```
<class 'set_iterator'>
```

```
In [216]: 1 print(type(iter({0: "0", 1: "1", 2: 2})))
```

```
<class 'dict_keyiterator'>
```

```
In [217]: 1 l = [4, 3, 2, 1, 0]
          2 iterator = iter(l)
          3 for i in iterator:
          4     print(i)
```

```
4
3
2
1
0
```

```
In [218]: 1 try:
          2     print(next(l))
          3 except TypeError as ex:
          4     print(type(ex).__name__, ":", *ex.args, ".", sep = " ")
```

```
TypeError: 'list' object is not an iterator.
```

```
In [219]: 1 start = time.perf_counter()
          2 squaresGenerator = (i ** 10 for i in range(1_000_000))
          3 end = time.perf_counter()
          4 print(f"Elapsed: {end - start}")
```

```
Elapsed: 0.00015254699997058196
```

```
In [220]: 1 start = time.perf_counter()
          2 squaresListComp = [i ** 10 for i in range(1_000_000)]
          3 end = time.perf_counter()
          4 print(f"Elapsed: {end - start}")
```

```
Elapsed: 0.21923032399990916
```

```
In [221]: 1 print((7.16e-5 / 1.824e-1) ** -1)
          2 print(105 / 9)
          3 print(10.5e1 / 9e0)
          4 print(99 / 1)
          5 print(9.9e1 / 1e0) # 1 order of magnitude difference means a multiple of
                           between 10 and 100
```

```
2547.486033519553
11.666666666666666
11.666666666666666
99.0
99.0
```

```
In [222]: 1 print(999_999 ** 10)
```

```
999990000004499980000209999748000209999800000044999990000001
```

```
In [223]: 1 print(squaresGenerator is iter(squaresGenerator))
```

```
True
```

```
In [224]: 1 l = [10, "abc", 3.14, True]
          2 for i, e in enumerate(l):
```

```
3 print(f"{i}: {e}")
```

```
0: 10  
1: abc  
2: 3.14  
3: True
```

```
In [225]: 1 def f(a, *b, x, **y):  
2         print("Positional/Named:", a)  
3         print("Variable positional:", b)  
4         print("Named only:", x)  
5         print("Variable named:", y)  
6  
7  
8 f(0, 1, 2, x = 3, y = 4, z = 5)
```

```
Positional/Named: 0  
Variable positional: (1, 2)  
Named only: 3  
Variable named: {'y': 4, 'z': 5}
```

```
In [226]: 1 f(10, 2, x = 10, 20)
```

```
Cell In[226], line 1  
f(10, 2, x = 10, 20)  
      ^
```

SyntaxError: positional argument follows keyword argument

```
In [227]: 1 f(10, 2, x = 10, y = 20)
```

```
Positional/Named: 10  
Variable positional: (2,)  
Named only: 10  
Variable named: {'y': 20}
```

```
In [228]: 1 f(10, z = 2, x = 10, y = 20)
```

```
Positional/Named: 10  
Variable positional: ()  
Named only: 10  
Variable named: {'z': 2, 'y': 20}
```

```
In [229]: 1 f(10, x = 2, y = 10, z = 20)
```

```
Positional/Named: 10  
Variable positional: ()  
Named only: 2  
Variable named: {'y': 10, 'z': 20}
```

```
In [230]: 1 f(a = 10, 2, x = 10, y = 20)
```

```
Cell In[230], line 1  
f(a = 10, 2, x = 10, y = 20)  
      ^
```

SyntaxError: positional argument follows keyword argument

```
In [231]: 1 f(x = 10, y = 20, z = 30, a = 10)
```

```
Positional/Named: 10
Variable positional: ()
Named only: 10
Variable named: {'y': 20, 'z': 30}
```

```
In [232]: 1 try:
          2     f(10, a = 2, x = 10, y = 20)
          3 except TypeError as ex:
          4     print(type(ex).)
```

TypeError

```
In [233]: 1 def f1(a, *, x, **y):
          2     print(f"a: {a}, x: {x}, y: {y}")
```

```
In [234]: 1 try:
          2     f1(10, a = 20, x = 2, y = 10)
          3 except TypeError as ex:
          4     print(type(ex).)
```

TypeError

```
In [235]: 1 f1(a = 1, x = 2)
```

a: 1, x: 2, y: {}

```
In [236]: 1 print(max("cba"))
```

c

```
In [237]: 1 def f2(a, b):
          2     print(f"a: {a}; b: {b}")
          3
          4
          5 f2(2, 3)
```

a: 2; b: 3

```
In [238]: 1 f2(2, b = 3)
```

a: 2; b: 3

```
In [239]: 1 f2(a = 2, 3)
```

Cell In[239], line 1

```
f2(a = 2, 3)
```

^

SyntaxError: positional argument follows keyword argument

```
In [240]: 1 f2(b = 1, a = 0)
```

a: 0; b: 1

```
In [241]: 1 def f3(a = 0, b):
          2     print(f"a: {a}; b: {b}")
```

Cell In[241], line 1

```
def f3(a = 0, b):  
    ^
```

SyntaxError: parameter without a default follows parameter with a default

```
In [242]: 1 def f3(a, b = 0):  
2         print(f"a: {a}; b: {b}")  
3  
4  
5 f3(-1, 1)
```

a: -1; b: 1

```
In [243]: 1 f3(a = -1, 1)
```

Cell In[243], line 1

```
f3(a = -1, 1)  
    ^
```

SyntaxError: positional argument follows keyword argument

```
In [244]: 1 f3(a = -1, b = 1)
```

a: -1; b: 1

```
In [245]: 1 def f4(a, b = 1, *, x):  
2         print(a, b, x)  
3  
4  
5 f4(0, 2, x = 3)
```

0 2 3

```
In [246]: 1 def f5(a, b = 1, *, x = 2, y):  
2         print(a, b, x, y)  
3  
4  
5 f5(0, 2, x = 3, y = 4)
```

0 2 3 4

```
In [247]: 1 def f6(**x):  
2         print(x)  
3  
4  
5 f6(a = 1, b = 2, c = 3)
```

{'a': 1, 'b': 2, 'c': 3}

```
In [248]: 1 def f7(*, **x):  
2         print(x)  
3  
4  
5 f7(a = 1, b = 2)
```

Cell In[248], line 1

```
def f7(*, **x):  
    ^
```

SyntaxError: named arguments must follow bare *

```
In [249]: 1 def f8(a, *, b):  
2         print(f"a: {a}; b: {b}")  
3  
4  
5 f8(b = 1, a = 0)
```

a: 0; b: 1

```
In [250]: 1 def toJSON(a, *, b, **c):  
2         print(a, b, c)  
3  
4  
5 toJSON(0, b = 3, c1 = 0, c2 = 1)
```

0 3 {'c1': 0, 'c2': 1}

```
In [251]: 1 def avg(a, *b):  
2         return (sum(b) + a) / (len(b) + 1)  
3  
4  
5 print(avg(0, 1, 2, 3, 4))  
6 print(avg(0))  
7 print(avg(0, 1))
```

2.0

0.0

0.5

```
In [252]: 1 def separator(b = 10, *, a = "-"):  
2         return a * b  
3  
4  
5 print(separator())  
6 print(separator(a = "A"))  
7 print(separator(5))  
8 print(separator(a = "B", b = 5))  
9 print(separator(5, a = "B"))
```

AAAAAAAAAA

BBBBBB

BBBBBB

```
In [253]: 1 print(sep = "|", *("AB", "CD"))
```

AB|CD

```
In [254]: 1 def f(a, *b, c):  
2         print(a, b, c)
```

```
3
4
5 f(c = 2, 0, 1, 2)
```

Cell In[254], line 5

```
f(c = 2, 0, 1, 2)
      ^
```

SyntaxError: positional argument follows keyword argument

```
In [255]: 1 def f(*a, b = 0):
          2     print(a, b)
          3
          4
          5 f(b = 1, 0, 2)
```

Cell In[255], line 5

```
f(b = 1, 0, 2)
      ^
```

SyntaxError: positional argument follows keyword argument

```
In [256]: 1 try:
          2     f(b = 1, *(0, 2))
          3 except TypeError as ex:
          4     print(type(ex).      )
```

TypeError

```
In [257]: 1 def f(a, *, b):
          2     print(a, b)
          3
          4
          5 f(b = 1, 0)
```

Cell In[257], line 5

```
f(b = 1, 0)
      ^
```

SyntaxError: positional argument follows keyword argument

```
In [258]: 1 def f(a, b):
          2     print(a, b)
          3
          4
          5 f(b = 1, 0)
```

Cell In[258], line 5

```
f(b = 1, 0)
      ^
```

SyntaxError: positional argument follows keyword argument

```
In [259]: 1 def f(a, b = 2):
          2     print(a, b)
          3
          4
          5 f(b = 1, 0)
```

Cell In[259], line 5

```
f(b = 1, 0)
```

^

SyntaxError: positional argument follows keyword argument

In [260]:

```
1 def f(a, b = 2):
2     print(a, b)
3
4
5 f(b = 1, *(0,))
```

0 1

In [261]:

```
1 anonymous = lambda iterable: len(set(iterable))
2 print(anonymous("abcd"))
3 print(anonymous("abab"))
4 try:
5     print(anonymous())
6 except TypeError as ex:
7     print(type(ex).      )
8 print(anonymous(""))
9 print(anonymous([0, 1, 2]))
10 print(anonymous([0]))
11 print(anonymous([]))
12 print(anonymous([1, 1, 2, 2, 3, 4]))
```

```
4
2
TypeError
0
3
1
0
4
```

In [262]:

```
1 s = "This is the first sentence. This is the scecond sentence. This is not the
2 fourth sentence, it is the third sentence."
3
4
5 def f(s = ""):
6     return dict(collections.Counter(s))
7
8
9 print(f(s))
print(f(""))
```

```
{'T': 3, 'h': 9, 'i': 10, 's': 13, ' ': 20, 't': 13, 'e': 17, 'f': 2, 'r': 3, 'n': 10, 'c': 6, 'l.': 3, 'o': 3, 'd': 2, 'u': 1, ',': 1}
{}
```

In [263]:

```
1 s = "This is the first sentence. This is the scecond sentence. This is not the
2 fourth sentence, it is the third sentence."
3
4
5 def f(s = ""):
```

```

3     words = re.split("\\.| |", s)
4     for word in words:
5         if word == "":
6             words.remove(word)
7     frequencies = dict.fromkeys(set(words), 0)
8     for word in words:
9         frequencies[word] += 1
10    print(frequencies)

```

```

11    f(s)

```

```

{'not': 1, 'is': 4, 'This': 3, 'it': 1, 'third': 1, 'fourth': 1, 'scecond': 1, 'first': 1, 'the': 4, 'sentence': 4}

```

```

In [264]: 1 def f(s = ""):
2         s = s.replace(".", "")
3         s = s.replace(",", "")
4         words = s.split()
5         return dict(collections.Counter(words))
6
7
8     print(f(s))

```

```

{'This': 3, 'is': 4, 'the': 4, 'first': 1, 'sentence': 4, 'scecond': 1, 'not': 1, 'fourth': 1, 'it': 1, 'third': 1}

```

```

In [265]: 1 print(round(1.812))

```

```

2

```

```

In [266]: 1 n, d = (0.3).as_integer_ratio()
2         print(n, d)
3         print(1 / 3)

```

```

5404319552844595 18014398509481984
0.3333333333333333

```

```

In [267]: 1 print(round(0.325, 2))

```

0.33

```
In [268]: 1 print(f"{0.325:.40f}")
```

0.32500000000000000111022302462515654042363

```
In [269]: 1 def toJSON(a, *, b, **c):
2           print(a, b, c)
3
4
5 try:
6     toJSON(0, 1, 2, b = 3, c1 = 4, c2 = 5)
7 except TypeError as ex:
8     print(type(ex).      )
```

TypeError

```
In [270]: 1 def toJSON(a, *, b, **c):
2           print(a, b, c)
3
4
5 toJSON(0, b = 3, c1 = 4, c2 = 5)
```

```
0 3 {'c1': 4, 'c2': 5}
```

```
In [271]: 1 def toJSON(a, *):
           2     print(a)
           3
           4
           5 toJSON(0, 1, 2)
```

Cell In[271], line 1

```
def toJSON(a, *):
```

```
SyntaxError: named arguments must follow bare *
```

```
In [272]: 1 def toJSON(a, *args):
           2     print(a, args)
           3
           4
           5 toJSON(0, 1, 2)
```

 $\theta \in (1, 2)$

```
In [273]: 1 l = ["AB CD", "EF CC", "GH BC", "IJ AC", "KL CA", "MN JQ", "OP BA"]
          2 print(sorted(l, key = lambda e: (e[3], e[4])))
```

```
['IJ AC', 'OP BA', 'GH BC', 'KL CA', 'EF CC', 'AB CD', 'MN JQ']
```

```
In [274]: 1 # print(sorted(l, key = lambda e: e[len(e) - i] for i in range(len(e))))
          2 print(sorted(l, key = lambda e: e[len(e) - 1]))
```

```
['KL CA', 'OP BA', 'EF CC', 'GH BC', 'IJ AC', 'AB CD', 'MN JQ']
```

```
In [275]: 1 a = (1 for _ in range(5))
          2 print(a)
          3 print(lambda _: "ABCD")
          4 print(sorted)
          5 print(range)
```

```
<generator object <genexpr> at 0x72d303ea9f20>
<function <lambda> at 0x72d2ffef83b0>
<built-in function sorted>
<class 'range'>
```

```
In [276]: 1 def f(*a, b):
          2     print(a, b)
          3
          4
          5 f(b = 0, *(1, 2))
```

```
(1, 2) 0
```

```
In [277]: 1 def f(a, *b, c):
          2     print(a, b, c)
          3
          4
          5 f(0, c = 1, *(2, 3))
```

```
0 (2, 3) 1
```

```
In [278]: 1 print(statistics.fmean)
```

```
<function fmean at 0x72d348406f00>
```

```
In [279]: 1 print(dict([("a", 97), ("b", 98), ("c", 99)]))
```

```
{'a': 97, 'b': 98, 'c': 99}
```

```
In [280]: 1 widgets = [f"w{i}" for i in range(1, 21)]
          2 skus = [f"sku{i}" for i in range(1, len(widgets) + 1)]
          3 print(widgets)
          4 print(skus)
```

```
['w1', 'w2', 'w3', 'w4', 'w5', 'w6', 'w7', 'w8', 'w9', 'w10', 'w11', 'w12', 'w13', 'w14', 'w15',
'w16', 'w17', 'w18', 'w19', 'w20']
['sku1', 'sku2', 'sku3', 'sku4', 'sku5', 'sku6', 'sku7', 'sku8', 'sku9', 'sku10', 'sku11', 'sku12',
'sku13', 'sku14', 'sku15', 'sku16', 'sku17', 'sku18', 'sku19', 'sku20']
```

```
In [281]: 1 print({w: s for w, s in zip(widgets, skus)})
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',
'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':
'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':
'sku20'}
```

```
In [282]: 1 print(dict(zip(widgets, skus)))
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',  
'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':  
'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':  
'sku20'}
```

```
In [283]: 1 suits = "shdc" # Spades, Hearts, Diamonds, Clubs  
2 ranks = list("23456789") + ["10", "J", "Q", "K", "A"]  
3 print(suits)  
4 print(ranks)
```

```
shdc  
['2', '3', '4', '5', '6', '7', '8', '9', '10', 'J', 'Q', 'K', 'A']
```

```
In [284]: 1 print([(rank, suit) for rank in ranks] for suit in suits))
```

```
[(('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'),  
(('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's'))], (('2', 'h'), ('3', 'h'), ('4', 'h'),  
(('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'),  
(('K', 'h'), ('A', 'h'))], (('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'),  
(('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd'))], (('2', 'c'),  
(('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'),  
(('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c'))]
```

```
In [285]: 1 pprint.pprint([(rank, suit) for rank in ranks] for suit in suits))
```

```
[('2', 's'),  
 ('3', 's'),  
 ('4', 's'),  
 ('5', 's'),  
 ('6', 's'),  
 ('7', 's'),  
 ('8', 's'),  
 ('9', 's'),  
 ('10', 's'),  
 ('J', 's'),  
 ('Q', 's'),  
 ('K', 's'),  
 ('A', 's')],  
 [('2', 'h'),  
 ('3', 'h'),  
 ('4', 'h'),  
 ('5', 'h'),  
 ('6', 'h'),  
 ('7', 'h'),  
 ('8', 'h'),  
 ('9', 'h'),  
 ('10', 'h'),  
 ('J', 'h'),  
 ('Q', 'h'),  
 ('K', 'h'),  
 ('A', 'h')],  
 [('2', 'd'),  
 ('3', 'd'),  
 ('4', 'd'),  
 ('5', 'd'),  
 ('6', 'd'),  
 ('7', 'd'),  
 ('8', 'd'),  
 ('9', 'd'),  
 ('10', 'd'),  
 ('J', 'd'),  
 ('Q', 'd'),  
 ('K', 'd'),  
 ('A', 'd')],  
 [('2', 'c'),  
 ('3', 'c'),  
 ('4', 'c'),  
 ('5', 'c'),  
 ('6', 'c'),  
 ('7', 'c'),  
 ('8', 'c'),  
 ('9', 'c'),  
 ('10', 'c'),  
 ('J', 'c'),  
 ('Q', 'c'),  
 ('K', 'c'),  
 ('A', 'c')]]
```

```
In [286]: 1 print([list(zip(ranks, s * 13)) for s in suits])
```

```
[(('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'), ('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's'))], [(('2', 'h'), ('3', 'h'), ('4', 'h'), ('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'), ('K', 'h'), ('A', 'h'))], [(('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'), ('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd'))], [(('2', 'c'), ('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'), ('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c'))]
```

In [287]:

```
1 def f(list1, *, reverse = False):
2     if not reverse:
3         return sorted(list1)
4     else:
5         return sorted(list1, reverse = True)
6
7
8 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
9 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
10 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
11 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

In [288]:

```
1 def f(list1, *, reverse = False):
2     return sorted(list1, reverse = reverse)
3
4
5 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
6 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
7 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
8 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

In [289]:

```
1 print(hex(ord(chr(0xE3).upper())))
```

0xc3

In [290]:

```
1 print("abc" in "defabcghi")
```

True

In [291]:

```
1 print("abc" in "abbde")
```

False

In [292]:

```
1 def solve(a, b, c):
2     solution = ((-1 * b) + math.sqrt((b * b) - (4 * a * c))) / (2 * a)
3     return round(solution, 2)
4
5
```

```
6 print(solve(3, 4, -5))
7 print(solve(1, -5, -8))
```

```
0.79
6.27
```

```
In [293]: 1 def encrypt(s):
          2     return "".join(chr(ord(c) + 10) for c in s)
          3
          4
          5 print(encrypt("ABCDEFGHIJKLMNOPQRSTUVWXYZ"))
```

```
KL MN OP QR ST UV WX YZ [\]^_`abcd
```

```
In [294]: 1 def decrypt(s):
          2     return "".join(chr(ord(c) - 10) for c in s)
          3
          4
          5 print(decrypt("KL MN OP"))
          6 print(decrypt("S}kkm*%o\x81~yx"))
```

```
ABCDEF
Isaac Newton
```

```
In [295]: 1 print(10 % 3)
```

```
1
```

```
In [296]: 1 a = array.array("l", [20])
          2 print(a)
```

```
array('l', [20])
```

```
In [297]: 1 for i in a:
          2     print(i)
```

```
20
```

```
In [298]: 1 print(sys.getsizeof(a))
```

```
88
```

```
In [299]: 1 try:
          2     print(sys.getallocatedblocks(a))
          3 except TypeError as e:
          4     print(type(e).      )
```

```
TypeError
```

```
In [300]: 1 print(sys.getsizeof(1))
```

```
28
```

```
In [301]: 1 def f(s):
          2     print(s)
          3
```

```

4
5 try:
6     f(*"hello")
7 except TypeError as ex:
8     print(type(ex).      , ex.args)

```

TypeError ('f()' takes 1 positional argument but 5 were given',)

```

In [302]: 1 def func(s):
2           print(s)
3
4
5 def apply(func, *str):
6     result = func(*str)
7     return result
8
9
10 apply(func, "john")

```

john

```

In [303]: 1 try:
2           apply(func, "john", "smith")
3 except TypeError as ex:
4     print(type(ex).      , *ex.args)

```

TypeError func() takes 1 positional argument but 2 were given

```

In [19]: 1 def encloseGetPower(b):
2           print("hex(id(b)):", hex(id(b)))
3
4           def getPower(e):
5               print("hex(id(e)):", hex(id(e)))
6
7               return b ** e
8
9           print("getPower:", getPower)
10
11          return getPower
12
13
14 getBinaryPower = encloseGetPower(2)
15 print(getBinaryPower(31))
16 getTertiaryPower = encloseGetPower(3)
17 print(getTertiaryPower(4))

```

```

hex(id(b)): 0xad7e50
getPower: <function encloseGetPower.<locals>.getPower at 0x7c0459d62610>
hex(id(e)): 0xad81f0
2147483648
hex(id(b)): 0xad7e70
getPower: <function encloseGetPower.<locals>.getPower at 0x7c0459d62560>
hex(id(e)): 0xad7e90
81

```

```
In [16]: 1 print("getBinaryPower:", getBinaryPower)
2 print("encloseGetPower:", encloseGetPower)
3 try:
4     print("getPower:", getPower)
5 except NameError as ex:
6     print(type(ex)., ex.args)
```

```
getBinaryPower: <function encloseGetPower.<locals>.getPower at 0x7c0459d62980>
encloseGetPower: <function encloseGetPower at 0x7c0459d62820>
NameError ("name 'getPower' is not defined",)
```

```
In [17]: 1 print("getBinaryPower:", getBinaryPower.
2 print("encloseGetPower:", encloseGetPower.
```

```
getBinaryPower: (<cell at 0x7c0459c0b3a0: int object at 0xad7e50>,)
encloseGetPower: None
```

```
In [307]: 1 def outer(a, b):
2         print(hex(id(a)))
3         sumTotal = a + b
4
5         def inner():
6             prod = a * b
7             print(a, b, sumTotal, prod)
8             return "You just called a closure!"
9
10        return inner
11
12
13 f = outer(2, 3)
```

```
0xad7e50
```

```
In [308]: 1 f = outer(2, 3)
```

```
0xad7e50
```

```
In [309]: 1 print(f)
```

```
<function outer.<locals>.inner at 0x72d2ffef8a90>
```

```
In [310]: 1 print(f.
)
```

```
(<cell at 0x72d30321edd0: int object at 0xad7e50>, <cell at 0x72d30321fa30: int object at 0xad7e70>,
<cell at 0x72d2fff00670: int object at 0xad7eb0>)
```

```
In [311]: 1 f()
```

```
2 3 5 6
```

```
Out[311]: 'You just called a closure!'
```

```
In [312]: 1 def execute(func):
2
3         def inner(a, b):
```

```

4         return func(a, b)
5
6     return inner
7
8
9 def add(a, b):
10     return a + b
11
12
13 add_executor = execute(add)
14 print(add_executor.

```

(<cell at 0x72d2fff01030: function object at 0x72d2ffef8b40>,)

In [313]: 1 print(hex(id(add)))

0x72d2ffef8b40

In [314]: 1 try:
2 print(hex(add))
3 except TypeError as ex:
4 print(type(ex).)

TypeError

In [315]: 1 add_executor(2, 3)

Out[315]: 5

In [316]: 1 def f():
2 pass
3
4
5 print(f.)

None

In [317]: 1 def f():
2
3 def g():
4 return
5
6 return g
7
8
9 h = f()
10 print(f.)
11 print(h.)

None

None

In [318]: 1 a = 0
2 try:

```
3     print(a.          )
4 except AttributeError as e:
5     print(type(e).    )
```

AttributeError

```
In [319]: 1 a = []
          2 try:
          3     print(a.          )
          4 except AttributeError as e:
          5     print(type(e).    )
```

AttributeError

```
In [320]: 1 def execute(func):
          2
          3     def inner(*posargs, **kwargs):
          4         return func(*posargs, **kwargs)
          5
          6     return inner
          7
          8
          9 def add(a, b, c):
         10     return a + b + c
         11
         12
        13 def say_hello(name, *, formal = True):
        14     return f"Pleased to meet you, {name}" if formal else f"Hi, {name}!"
        15
        16
        17 exec_add = execute(add)
        18 exec_greet = execute(say_hello)
```

```
In [321]: 1 exec_add(1, 2, 3)
```

Out[321]: 6

```
In [322]: 1 exec_greet("Michael", formal = False)
```

Out[322]: 'Hi, Michael!'

```
In [323]: 1 exec_greet("Michael", formal = True)
```

Out[323]: 'Pleased to meet you, Michael'

```
In [401]: 1 print((0.1).as_integer_ratio())
```

(3602879701896397, 36028797018963968)

```
In [325]: 1 print((0.125).as_integer_ratio())
```

(1, 8)

```
In [326]: 1 print((0.2).as_integer_ratio())
          2 print(f"{0.2:.40f}")
```

```
(3602879701896397, 18014398509481984)
0.20000000000000000111022302462515654042363
```

```
In [327]: 1 print(math.pi.as_integer_ratio())
          2 n, d = math.pi.as_integer_ratio()
          3 print(n / d)
```

```
(884279719003555, 281474976710656)
3.141592653589793
```

```
In [359]: 1 try:
          2     a1 = a1 + 1
          3     print(a1)
          4 except NameError as ex:
          5     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
NameError: name 'a1' is not defined.
```

```
In [329]: 1 def getDiagonalMatrix(val, rows, columns):
          2     diagonalMatrix = [[val if ce == r else 0 for ce in range(4)] for r in
          3     range(5)]
          4     return diagonalMatrix
```

```
In [352]: 1 def getDiagonalMatrix(rows, columns, *, val):
          2     diagonalMatrix = [[val if cl == row else 0 for cl in range(columns)]
          3     for row in range(rows)]
          4     return diagonalMatrix
```

```
In [330]: 1 def getFactorial(n):
          2     factorial = 1
          3     for i in range(1, n + 1):
          4         factorial *= i
          5     return factorial
```

```
In [331]: 1 result = getFactorial(5)
          2 print(result)
```

```
120
```

```
In [355]: 1 result = getDiagonalMatrix(3, 4, val = 5)
          2 pprint.pprint(result, width = 14)
```

```
[[5, 0, 0, 0],
 [0, 5, 0, 0],
 [0, 0, 5, 0]]
```

```
In [ ]: 1 print(f"{timeit.timeit('getFactorial(2)', globals = globals()):.4g}")
```

```
0.1713
```

In [393]:

```
1 n = 0.000000000000123456789
2 n = 4.482882e-05
3 exp = 0
4 while n < 1:
5     exp -= 3
6     n *= 1000
7 prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f", "a"
8 print(f"{n}{prefixes[abs(exp)//3]}")
```

44.82882μ

In []:

```
1 n = 1000000000
2 exp = 0
3 while n >= 1000:
4     exp += 3
5     n /= 1000
6 prefixes = "", "k", "M", "G", "T", "P", "E"
7 print(f"{n}{prefixes[exp//3]}")
```

1.0G

In [74]:

```
def toEnggNotation(n, scientificNotation = True, nSignificantDigits = 4):
1     exp = 0
2     oldN = n
3     if n > 1:
4         while n >= 1000:
5             exp += 3
6             n /= 1000
7             prefixes = "", "k", "M", "G", "T", "P", "E"
8             return f"{n:.3f}{prefixes[exp//3]}\t{oldN}" if
scientificNotation else f"{n}{prefixes[exp//3]}"
9         elif n < 1:
10            while n < 1:
11                exp -= 3
12                n *= 1000
13                prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f",
14                "a"
15            return f"{n:.{nSignificantDigits}f}
16            {prefixes[abs(exp)//3]}\t{oldN}" if scientificNotation else f"{n}
17            {prefixes[abs(exp)//3]}"
18        else:
19            return 1
20
21
22
23
24
25
26
```

```
1 start = time.perf_counter()
2 result = getFactorial(1_000)
3 end = time.perf_counter()
4 print(f"Result: {result}")
5 print(f"Elapsed: {end - start}")
```

Elapsed: 0.00015163200032475288

```
1 start = time.perf_counter()
2 result = getDiagonalMatrix(2, 4, val = 3)
3 end = time.perf_counter()
4 print(f"Result: {result}")
5 print(f"Elapsed: {end - start}")
```

Elapsed: 6.73889999234234e-05

```
1 def myTimer(f, *pos, **keyword):
2     start = time.perf_counter()
3     print(f(*pos, **keyword))
4     end = time.perf_counter()
5     return end - start
6
7
```

```
8 print(toEnggNotation(myTimer(getFactorial, 1000)))
9 print(toEnggNotation(myTimer(getDiagonalMatrix, 3, 5, val = 7)))
```

[illegible]

```

1 def encloseMyTimer(f):
2
3     def myTimer(*pos, **keyword):
4         start = time.perf_counter()
5         result = f(*pos, **keyword)
6         end = time.perf_counter()
7         print(toEnggNotation(end - start, True, 4))
8         return result
9
10    return myTimer
11
12
13 timed = encloseMyTimer(getDiagonalMatrix)
14 print(timed(2, 4, val = -5))
15 timed = encloseMyTimer(getFactorial)
16 print(timed(11))
17 print(encloseMyTimer(getDiagonalMatrix)(2, 4, val = -5))
18 print(encloseMyTimer(getFactorial)(11))

```

```
2.9860µ 2.986000254168175e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
1.7030µ 1.7029997252393514e-06
39916800
1.6840µ 1.684000380919315e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
982.0005n 9.820005288929678e-07
39916800
```

```
In [450]: 1 print(eval("2 * 3"))
          2 print(eval("x = 2 and print(x)"))
```

```
6
Traceback (most recent call last):

  File ~/src/PythonTestBed/venv/lib/python3.14/site-packages/IPython/core/interactiveshell.py:3748
in run_code
    exec(code_obj, self.user_global_ns, self.user_ns)

  Cell In[450], line 2
    print(eval("x = 2 and print(x)"))

  File <string>:1
    x = 2 and print(x)
      ^
SyntaxError: invalid syntax
```

```
In [452]: 1 eval("x = 2")
          2 print(x)
```

```
Traceback (most recent call last):

  File ~/src/PythonTestBed/venv/lib/python3.14/site-packages/IPython/core/interactiveshell.py:3748
in run_code
    exec(code_obj, self.user_global_ns, self.user_ns)

  Cell In[452], line 1
    eval("x = 2")

  File <string>:1
    x = 2
      ^
SyntaxError: invalid syntax
```

```
In [453]: 1 x = 2
          2 eval("x ** 2 - 1")
```

```
Out[453]: 3
```

```
In [20]: 1 def f(expr, x):
          2     print(eval(expr))
          3
          4
          5 x = 2
          6 f("x ** 2 - 1", x)
```

```
3
```

In [56]:

```
1 def f(x):
2     return True
3
4
5 try:
6     print(list(map(f(False), range(0, 5))))
7 except TypeError as ex:
8     print(type(ex).      + ":", ex.args[0])
```

TypeError: 'bool' object is not callable

In [67]:

```
1 def getExtremum(expr, interval, granularity, minimax):
2
3     def f(expr):
4
5         def g(x):
6             return eval(expr)
7
8         return g
9
10    evalExpr = f(expr)
11    print("Range:", list(range(interval[0], interval[1] + granularity,
12 granularity)))
13    if minimax == "minimum":
14        return min(map(evalExpr, range(interval[0], interval[1] +
15 granularity, granularity)))
16    elif minimax == "maximum":
17        return max(map(evalExpr, range(interval[0], interval[1] +
18 granularity, granularity)))
19    else:
20        return "Invalid minimax"
21
22    expr = "x ** 2 - 1"
23    print(getExtremum(expr, (-10, 10), 2, "minimum"))
24    print(getExtremum(expr, (-10, 10), 2, "maximum"))
25
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98
99
```

1
2
2

Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
-1
Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
99

In [60]:

```
1 expr = "x ** 2 - 1"
2 f = lambda x: eval(expr)
3 print(f(3))
```

8

In [30]:

```
def getExtremum(expr, interval, resolution, minimax):
    1
    2     def f(expr):
    3         def g(x):
    4             return eval(expr)
    5
    6         return g
    7
    8     evalExpr = f(expr)
    9     l = [interval[0] + (i * (interval[1] - interval[0]) / (resolution -
10 1)) for i in range(resolution)] # Cheated with the calc.
11     return min(map(evalExpr, l)) if minimax == "minimum" else
12 max(map(evalExpr, l))
13
14 expr = "x ** 2 - 1"
15 print(getExtremum(expr, (-10, 10), 5, "minimum"))
16 print(getExtremum(expr, (-10, 10), 5, "maximum"))
17 print(getExtremum(expr, (-10, 10), 6, "minimum"))
18 print(getExtremum(expr, (-10, 10), 6, "maximum"))
19 expr = "math.log10(x)"
20 print(getExtremum(expr, (1, 100), 20, "minimum"))
21 print(getExtremum(expr, (1, 100), 20, "maximum"))
22 expr = "math.sin(x)"
23 print(getExtremum(expr, (-10, 10), 20, "minimum"))
24 print(getExtremum(expr, (-10, 10), 20000, "maximum"))
25 expr = "x"
26 print(getExtremum(expr, (1, 6), 8, "minimum"))
27 print(getExtremum(expr, (1, 6), 8, "maximum"))
```

1
6
1
7
1
8
1
9

```
2
0
2
1
2
2
2
2
3
2
4
2
5
2
6
2
7
2
8
```

```
-1.0
99.0
3.0
99.0
0.0
2.0
-0.9999667804441291
0.999999996044897
1.0
6.0
```

In [80]:

```
def f(t):
    return t[0] ** t[1]

1
    return t[0] ** t[1]
2
l = [(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (3, 4), (4, 3),
3
    (4, 4)]

4
start = time.perf_counter()
for _ in range(10_000):
5
    result = []
    for t in l:
6
        result.append(f(t))
end = time.perf_counter()
7
forLoopTime = end - start
print(result)
8

start = time.perf_counter()
9
for _ in range(10_000):
1
    result = [f(t) for t in l]
0
end = time.perf_counter()
1
listCompTime = end - start
1
print(result)
1

2
start = time.perf_counter()
```

```

1  for _ in range(10_000):
3      result = list(map(f, l))
1  end = time.perf_counter()
4  mapTime = end - start
1  print(result)
5
1  print(toEnggNotation(forLoopTime), toEnggNotation(listCompTime),
6  toEnggNotation(mapTime))
1
7
1
8
1
9
2
0
2
1
2
2
2
3
2
4
2
5
2
6
2
7
2
8
2
9
3
0

```

```
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
```

```
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
```

```
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
```

```
10.0576m      0.010057615996629465 9.1698m      0.009169820994429756 7.2459m      0.007245856002555229
```

In []:

```

1  def power(x, y):
      return x ** y
2
3  def dist(pt1, pt2):
      return math.sqrt(sum(coord1 - coord2 for coord1, coord2 in zip(pt1,
4  pt2)))
5
5  def partial(f, operand2):

```

```

6         def calcPartialFunction(operand1):
7             return f(operand1, operand2)
8
9         return calcPartialFunction
10
11 squares = partial(power, 2)
12 print(squares(3))
13 print(squares(4))
14 dist_from_origin = partial(dist, (0, 0))
15 print(dist_from_origin((1, 1)))
16 gcd_13 = partial(math.gcd, 13)
17 log_2 = partial(math.log, 2)
18 log_10 = partial(math.log, 10)
19 log_16 = partial(math.log, 16)
20 print(log_2(8))
21 print(log_10(100))
22 print(log_16(65536))

```

```

9
16
1.4142135623730951
3.0
2.0
4.0

```

In [114]:

```
1 def norm(x, y):
2     return math.sqrt(x ** 2 + y ** 2)
3
4
5 def find_index_min(seq):
6     minima = min(seq)
7     return seq.index(minima)
8
9
10 def logged(f):
11
12     def g(*args):
13         print("Logging...")
14         start = time.perf_counter()
15         print(f(*args))
16         end = time.perf_counter()
17         print(toEnggNotation(end - start))
18         print("... Finished")
19
20     return g
21
22
23 loggedNorm = logged(norm)
24 loggedNorm(3, 4)
25 loggedFindIndexMin = logged(find_index_min)
26 loggedFindIndexMin([8, 7, 2, 9, 4])
```

```
Logging...
5.0
10.8400µ 1.0839998139999807e-05
... Finished
Logging...
2
6.8530µ 6.852998922113329e-06
... Finished
```

In [9]:

```
1 def encloseGetPower(f):
2
3     def getPower(*args):
4         return f(*args)
5
6     print(hex(id(getPower)))
7     return getPower
8
9
10 getPower = encloseGetPower(pow)
11 print(getPower(2, 5))
12 print(hex(id(getPower)))
13 print(hex(id(encloseGetPower)))
14 print(getPower. )
15 print(encloseGetPower. )
```

```
0x7606c01b5850
32
0x7606c01b5850
0x7606c01b5dd0
(<cell at 0x7606c017eb30: builtin_function_or_method object at 0x7606c7190630>,)
None
```

```
In [12]: 1 l = [(1, 2), (-4, -5.5), (10, -10), (-2, 2)]
        2 print(sorted(l, key = lambda t: abs(t[0] - t[1]), reverse = True))
```

```
[(10, -10), (-2, 2), (-4, -5.5), (1, 2)]
```

```
In [103]: 1 l = ["s", "0", "1"]
        2 card = "".join(reversed(l))
        3 print(card)
        4 card = ""
        5 for k in reversed(l):
        6     card += k
        7 print(card)
```

```
10s
10s
```

```
In [104]: suits = 'shdc' # Spades, Hearts, Diamonds, Clubs
        1 ranks = list('23456789') + ['10', 'J', 'Q', 'K', 'A']
        2 cards = [[rank + suit for rank in ranks] for suit in suits]

        3 def sorter(suits, reverse = False):
        4     return sorted(suits, key = lambda card:
        5         ranks.index(card.strip("s").strip("h").strip("d").strip("c")), reverse =
        6         reverse)

        7 sortedCards = []
        8 for suit in cards:
        9     sortedCards.append(sorter(suit, reverse = True))
        0 print(sortedCards)

        1 sortedCards = []
        2 for suit in cards:
        3     sortedCards.append(sorter(suit))
        4 print(sortedCards)
```

```
1
1
1
1
2
1
3
1
4
1
5
1
```

6
1
7
1
8

```
[[ 'As', 'Ks', 'Qs', 'Js', '10s', '9s', '8s', '7s', '6s', '5s', '4s', '3s', '2s'], [ 'Ah', 'Kh', 'Qh',  
'Jh', '10h', '9h', '8h', '7h', '6h', '5h', '4h', '3h', '2h'], [ 'Ad', 'Kd', 'Qd', 'Jd', '10d', '9d',  
'8d', '7d', '6d', '5d', '4d', '3d', '2d'], [ 'Ac', 'Kc', 'Qc', 'Jc', '10c', '9c', '8c', '7c', '6c',  
'5c', '4c', '3c', '2c']]  
[[ '2s', '3s', '4s', '5s', '6s', '7s', '8s', '9s', '10s', 'Js', 'Qs', 'Ks', 'As'], [ '2h', '3h', '4h',  
'5h', '6h', '7h', '8h', '9h', '10h', 'Jh', 'Qh', 'Kh', 'Ah'], [ '2d', '3d', '4d', '5d', '6d', '7d',  
'8d', '9d', '10d', 'Jd', 'Qd', 'Kd', 'Ad'], [ '2c', '3c', '4c', '5c', '6c', '7c', '8c', '9c', '10c',  
'Jc', 'Qc', 'Kc', 'Ac']]
```

In [141]:

1
2
3
4
5
6
7
8
9
1
0

```
data = {'S1': (100, 200, 80, 180), 'S2': (10, 20, 8, 18), 'S3': (50, 150, 50,  
150)}  
  
def findMinima(data, reverse = False):  
    minimax = min if reverse else max  
    return minimax(data, key = lambda d: data[d][1] - data[d][2])  
  
print(findMinima(data, reverse = True))  
print(findMinima(data))
```

S2
S1

In [166]:

1
2
3
4
5
6
7
8
9

```
quotes = [('AACC', 6.05, 6.07, 6.03, 6.05, 65800), ('AAME', 1.7, 1.82, 1.7,  
1.82, 4300), ('AON', 24.98, 25.07, 24.9, 24.94, 28200), ('AAPL', 317.99,  
319.57, 316.75, 317.13, 12901800), ('AATI', 3.82, 3.82, 3.74, 3.79, 194600),  
( 'AAWW', 60.89, 61.44, 60.5, 61.19, 272800), ('AAXJ', 65.4, 65.71, 65.28,  
65.56, 390300), ('ABAT', 4.01, 4.01, 3.95, 3.99, 656300), ('ABAX', 25.26,  
25.49, 25.04, 25.42, 73700), ('ABBC', 11.75, 11.88, 11.48, 11.53, 29700),  
( 'ABCB', 9.3, 9.3, 9.06, 9.14, 42600), ('ABCD', 3.25, 3.25, 3.11, 3.22,  
122800), ('ABCO', 48.75, 50.41, 46.9, 50.37, 66300), ('ABCW', 0.52, 0.61, 0.52,  
0.53, 83000), ('ABFS', 25.98, 26.27, 25.41, 25.5, 384900), ('ABIO', 3.96, 4,  
3.88, 4, 38500), ('ABMD', 11.94, 12, 11.69, 11.87, 122600), ('ABTL', 0.82,  
0.84, 0.82, 0.83, 28700), ('ABVA', 3.09, 3.25, 3.09, 3.25, 6200), ('ACAD',  
0.76, 0.76, 0.7, 0.74, 341500), ('ACAS', 7.52, 7.72, 7.52, 7.66, 5199800),  
( 'ACAT', 14.44, 14.44, 14.04, 14.2, 51700), ('ACCL', 8.11, 8.21, 7.94, 8.1,  
456100), ('ACET', 8.01, 8.04, 7.13, 7.73, 575600), ('ACFC', 1.69, 1.7, 1.5,  
1.6, 12300), ('ACFN', 3.82, 4, 3.82, 3.98, 53700), ('ACGL', 89.76, 90.14,
```

```
89.39, 89.92, 240900), ('ACGY', 22.41, 22.56, 22.25, 22.46, 86800), ('ACHN',
3.12, 3.2, 3.07, 3.16, 113700), ('ACTIW', 26.96, 27.03, 26.63, 26.8, 157000),
('ACLI', 33.65, 33.77, 33.45, 33.63, 28700), ('ACLS', 2.47, 2.63, 2.46, 2.53,
1818800), ('ACMR', 2.69, 2.84, 2.37, 2.71, 158600), ('ACOM', 25.2, 26.6, 24.9,
26.56, 265300), ('ACOR', 26.67, 27.07, 26.38, 27.04, 1415000), ('ACPW', 1.84,
1.89, 1.77, 1.85, 565500), ('ACTG', 27.2, 27.43, 26.86, 27.18, 228800),
('ACTI', 3.25, 3.26, 3.25, 3.26, 148500), ('ACTS', 2.08, 2.09, 2.07, 2.07,
130500), ('ACUR', 2.6, 2.64, 2.51, 2.6, 16000), ('ACWI', 46.53, 46.7, 46.32,
46.51, 286200), ('ACWX', 44.49, 44.66, 44.36, 44.6, 55500), ('ACXM', 18, 18.07,
17.81, 18.01, 289800), ('ADAM', 7.34, 7.49, 7.33, 7.44, 81700), ('ADAT', 0.6,
0.68, 0.59, 0.66, 86400), ('ADBE', 29.43, 29.71, 29.07, 29.14, 7585300),
('ADCT', 12.68, 12.69, 12.66, 12.68, 1660500), ('ADEP', 6.14, 6.14, 4.95, 5.61,
71000), ('ADES', 6.2, 6.22, 6, 6.19, 4800), ('ADGF', 4.31, 4.55, 4.31, 4.54,
10200)]
```

```
def myFilter(quote, percentage = 10):
    pass
```

```
percentage = 10
print(list(filter(lambda t: abs((quote[4] - quote[2]) / quote[2]) > percentage,
quotes)))
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[166], line 9
      5     pass
      6
      7
      8 percentage = 10
----> 9 print(list(filter(lambda t: abs((quote[4] - quote[2]) / quote[2]) > percentage, quotes)))

Cell In[166], line 9, in <lambda>(t)
----> 9 print(list(filter(lambda t: abs((quote[4] - quote[2]) / quote[2]) > percentage, quotes)))

NameError: name 'quote' is not defined
```

In []:

1